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## Solar Lighting System

I need the circuit of a solar lighting system without solar panel. Where can I purchase it from?

Sujit George  
Bengaluru

**EFY:** The complete kit of the mentioned circuit is available from EFY associate Kits'n'Spares. For details, you may visit their website [www.kitsnspares.com](http://www.kitsnspares.com)

## SMD Rework Stations

I thank Col. (Retd) N.C. Pande for answering the query on OPC drum and toner under Q&A section. I would also like to thank Tularam Madgaji Bansod for his article 'SD/MMC Memory Card Fundamentals' and M. Balaji Prasad for his article 'Creating and Sustaining an Effective e-waste Management System' in June issue.

I also request EFY to cover SMD rework stations in Buyers' Guide section.

Subhasis Sahoo  
Electronics Lab,  
SMS-II, RSP, SAIL  
Rourkela, Orissa

## Circuits for Publication

I am doing M.Tech in power electronics. I have a circuit and want to send it to EFY for publication. Please tell me the address and the procedure for submitting my circuit.

Teja  
Vijayawada

☐ I want to send some circuit ideas for publication in EFY. Please tell me the criteria for submission and the format of the article.

Paras Raval  
Rajkot

**EFY:** You may please send the details to "The Editor, EFY Enterprises Pvt Ltd, D-87/1, Okhla Industrial Area Phase I, New Delhi - 110020." For guidelines to submit, visit the link [www.electronicsforu.com/electronics-foru/articles/WI-EDT.pdf](http://www.electronicsforu.com/electronics-foru/articles/WI-EDT.pdf)

## Voltage Stabiliser

In the 'Voltage Stabiliser Using PIC-16F877A' construction project published in August issue, I have the following doubts:

1. There is a printing error in Fig. 2

where the stabilised output is shown. It should be from the other end of the secondary winding of the buck-boost transformer. Otherwise, the input equals the output.

2. As there is no overload protection, the MOSFETs may fail on accidental short-circuit in the output.

3. The circuit is based on input supply sensing only, so it would not know about any abnormal change taking place at the output, say, voltage drop in the secondary winding owing to the load. Can it be designed in output-sensing mode?

4. For buck one MOSFET conducts, and for boost the other MOSFET conducts. And the path for boost is, say, when L is positive with respect to N in the first half cycle, D5, P, bottom half of primary X2, drain of T2, source of T2, COM, D6 and then N. And for N positive with respect to L in the next half cycle, it takes the path from D7, P, bottom half of primary X2, drain of T2, source of T2, COM, D4 and then to L. Similarly, the path for buck uses T1 and the upper half of primary X2 with the help of bridge diodes D4 through D7.

If this be the case, the load current (or the primary current) always passes through bridge diodes D4 through D7, which are rated only one ampere each. Will it not restrict the output VA handling capacity?

5. When the input falls below 172V (say, 130V), the DC output from 9V-0.9V transformer for the circuit will go below the minimum required voltage of 8V for the 7805 to maintain the regulated output of 5V, which may result in improper functioning of the microcontroller.

Uma Charan Patnaik  
Hyderabad

**The author Prof. K. Padmanabhan replies:**

1. The output has to be taken from the bottom of the secondary and not the top as shown. Thanks for pointing out the mistake.

2. Accidental short of the stabilised output is unlikely. It is only as likely as shorting the mains supply accidentally. The fuse will blow off first. The MOSFET will not fail because its contribu-